

MADHU PRAKASH S

Mechanical Design Engineer

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🌐 linkedin.com/in/madhuprakashs 🔗 Portfolio LInk



Professional Summary

Mechanical Design Engineer with an M.Tech *in Mechanical Systems (Robotics & AI)* and a B.E. in Mechanical Engineering, with hands-on experience in 3D CAD modelling, automation fixture design, manufacturing drawings, and mechanical assembly development. Experienced in designing custom mechanical components and integrating them with standard industrial components for automation applications. Familiar with GD&T, tolerance analysis, DFM/DFA, manufacturing processes, and industrial robotics. Proficient in Autodesk Fusion 360, with hands-on exposure to CATIA V5, AutoCAD, and SolidWorks.

Internship

Jan 2026 – Present
Bengaluru, India

Intern under CPED(Central Production and Engineering Department)

Wipro Hydraulics (Wipro Infrastructure Engineering)

- Analyzed 9 manual operations in a hydraulic cylinder sub-assembly line and identified automation opportunities across 5 key activities.
- Developed CAD concepts for automation solutions including grippers, wiper blade mechanisms, and a trunnion tray fixture.
- Prepared 2D manufacturing drawings with basic GD&T specifications for designed components.
- Proposed automation concepts focused on reducing manual effort, improving process consistency, and increasing assembly efficiency.
- Contributed to evaluation and shortlisting of 1 automation concept for further consideration.

Nov 2024 – Dec 2024
Bengaluru, India

Industrial Robotics Training

GTTC (Government Tool Room & Training Centre)

- Trained in robot operations, path planning, and trajectory tracing specifically for **KUKA Industrial Robots**.
- **Impact:** Gained hands-on experience in programming robotic systems for precision movement and industrial task execution.

Education

2023 – 2025
Bengaluru, India

Master of Technology in Robotics and Artificial Intelligence

Ramaiah Institute of Technology

8.97 cgpa

2019 – 2023
Bengaluru, India

Bachelor of Engineering in Mechanical

Dr. Ambedkar Institute Of Technology

8.0 cgpa.

Projects

Jan 2025 – Jul 2025

AI-Powered Handheld Thermal Imaging Device For Real-Time Human, Detection In Disaster Response Operations

PG Project

- Developed a helmet-mounted thermal camera system for real-time human detection in disaster response operations.
- Implemented detection using **Python, OpenCV, and YOLOv8 object detection** to identify victims through heat signatures, transmitting live data to remote command centers.

- Designed the protective enclosure/mounting housing for the thermal camera and microcontroller using Autodesk Fusion 360 for secure helmet integration and field usability.

Impact: Secured **project funding** from the Alumni Association and successfully validated the system with performance results of **78% mean accuracy, 82% mean recall, and 78% mean precision**.

Nov 2022 – Jul 2023

Fabrication of Small Segment Car Operated by Electric Power

UG Major Project/VTU

- Led team of four members in end-to-end execution of foldable electric vehicle project, coordinating task allocation for design, fabrication, and final assembly.
- Designed vehicle body and chassis using AutoCAD and Autodesk Fusion 360; performed load calculations with Factor of Safety of 2 to establish maximum load capacity of 80 kg.
- Coordinated fabrication using 3 mm mild steel sheet and MS square tubes, supported welding and assembly, and conducted physical load validation testing to confirm structural integrity.
- Developed fully functional prototype with foldable form factor, demonstrating practical mobility and compact storage capability.

Impact: Successfully developed a fully functional prototype that could fold into a suitcase-like form factor, demonstrating practical mobility, compact storage, and project feasibility.

2026 – Present

Self-Initiated CAD Modelling Portfolio

Fusion 360 & CATIA V5 | 2026 – Present

- **Developed 10+ mechanical parts and surface models**, including complex components, sheet-metal designs, and aerospace-oriented models.
- **Designed 5 multi-component assemblies in CATIA V5**, including Butterfly Valve, Double Bearing, Motor Blower, Press Tool, and Pulley System assemblies.
- Applied **parametric modelling, assembly design, surface modelling, sheet-metal modelling, and engineering drawing practices**.
- <https://maddy0330.github.io/portfolio>  **click here to view my portfolio.**

Technical Skills

- **CAD & Design:** Autodesk Fusion 360, Catia V5, AutoCAD, SolidWorks, 3D part and assembly modelling, engineering drawings, GD&T, tolerance analysis, DFM/DFA
- **Manufacturing & Fabrication:** CNC programming (G-code/M-code), welding, mechanical assembly, load analysis and validation
- **Engineering & Process:** Mechanical design principles, manufacturing considerations, process optimization, problem-solving
- **Programming & Tools:** Python, VS Code, Google Colab
- **Quality & Operation:** Six Sigma and Lean Management

Certifications

Catia V5 Dassault Certification

Catia V5 Dassault Certification

Six Sigma and Lean Management

On-going training related to six sigma and lean management from CBEMG training institute

Declaration

I hereby declare that all information furnished above is authentic and accurate to the best of my belief.

Madhu Prakash S